

Central and Local Limit Theorems Applied to Asymptotic Enumeration

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Communicated by the Managing Editors

Received April 7, 1971

Let a double sequence $a_n(k) \geq 0$ be given. We prove a simple theorem on generating functions which can be used to establish the asymptotic normality of $a_n(k)$ as a function of k . Next we turn our attention to local limit theorems in order to obtain asymptotic formulas for $a_n(k)$. Applications include constant coefficient recursions, Stirling numbers, and Eulerian numbers.

1. INTRODUCTION

With a double sequence of non-negative numbers $a_n(k)$ we associate the normalized double sequence

$$p_n(k) = \frac{a_n(k)}{\sum_j a_n(j)}.$$

We say that $a_n(k)$ is asymptotically normal with mean μ_n and variance σ_n^2 if

$$\limsup_{n \rightarrow \infty} \sup_{x \in S} \left| \sum_{k \leq \sigma_n x + \mu_n} p_n(k) - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt \right| = 0. \quad (1.1)$$

Equivalently we say $a_n(k)$ satisfies a central limit theorem. If for *some* set S of real numbers

$$\limsup_{n \rightarrow \infty} \sup_{x \in S} \left| \sigma_n p_n([\sigma_n x + \mu_n]) - \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \right| = 0, \quad (1.2)$$

we say that $a_n(k)$ satisfies a local limit theorem on S . In view of the uniform continuity of $e^{-x^2/2}$ and Theorem 1 of [3, Section 9], condition (1.1) is equivalent to pointwise convergence.

If one can prove a central limit theorem for a sequence $a_n(k)$ of numbers arising in enumeration, then one has a qualitative feel for their behavior. A local limit theorem is better because it provides asymptotic information about $a_n(k)$ for

$$|k - \mu_n| = O(\sigma_n). \quad (1.3)$$

Unfortunately, the step from (1.1) to (1.2) is often not easy. Nevertheless, if (1.1) holds we can usually be fairly certain that (1.2) holds as well. The reason for this is quite simple:

$$(1.1) + (\text{"smoothness" of } a_n(k)) \Rightarrow (1.2).$$

For asymptotic purposes the range of k given by (1.3) is quite limited. However, since the mean of $r^k a_n(k)$ varies with r , it is possible to extend the range of the asymptotic formula.

The importance of these ideas is their simplicity, not their accuracy. A detailed study of the generating function can be expected to give more precise results. Consequently we will not discuss error estimates. In principle, the proofs lead to such estimates, but the calculations would be quite tedious.

We will begin by proving a central limit theorem based on the nature of the generating function $\sum a_n(k) z^n w^k$. Several applications will then be given. Next we turn our attention to local analogs of this theorem. We also state precisely the local limit theorem hinted at by Harper [5]. These results will be applied to the derivation of asymptotic formulas. Finally we discuss some open questions.

2. A CENTRAL LIMIT THEOREM

We now prove

THEOREM 1. *Let $f(z, w)$ have a power series expansion*

$$f(z, w) = \sum_{n, k \geq 0} a_n(k) z^n w^k$$

with non-negative coefficients. Suppose there exists (i) an $A(s)$ continuous and non-zero near 0, (ii) an $r(s)$ with bounded third derivative near 0, (iii) a non-negative integer m , and (iv) $\epsilon, \delta > 0$ such that

$$\left(1 - \frac{z}{r(s)}\right)^m f(z, e^s) - \frac{A(s)}{1 - z/r(s)} \quad (2.1)$$

is analytic and bounded for

$$|s| < \epsilon \quad |z| < |r(0)| + \delta.$$

Define

$$\mu = -\frac{r'(0)}{r(0)} \quad \sigma^2 = \mu^2 - \frac{r''(0)}{r(0)}. \quad (2.2)$$

If $\sigma \neq 0$, then (1) holds with $\mu_n = n\mu$ and $\sigma_n^2 = n\sigma^2$.

Proof. Let $f(z, e^s) = \sum \varphi_n(s) z^n$. By standard contour integration using the contour $|z| = |r(0)| + 2\delta/3$, we have for some constant C that the coefficient of z^n in (2.1) is less in absolute value than $C(|r(0)| + 2\delta/3)^{-n}$. Thus we have that the coefficient of z^n in

$$f(z, e^s) - \frac{A(s)}{(1 - z/r(s))^{m+1}}$$

is less in absolute value than

$$\sum_{k=0}^n C(|r(0)| + 2\delta/3)^{k-n} \binom{-m}{k} (-|r(s)|)^{-k}.$$

Provided $|r(s) - r(0)| \leq \delta/3$ we have

$$\begin{aligned} & \left| \varphi_n(s) - A(s) \binom{n+m}{m} r(s)^{-n} \right| \\ &= C \binom{n+m-1}{m-1} |r(s)|^{-n} \sum_{k=0}^n (1 + \delta/3 |r(s)|)^{k-n}. \end{aligned}$$

Since the sum is absolutely bounded for small s and

$$\binom{n+m-1}{m-1} = \frac{m}{n+m} \binom{n+m}{m},$$

it follows that

$$\varphi_n(s) = \binom{n+m}{m} r(s)^{-n} A(s) \left(1 + O\left(\frac{1}{n}\right)\right) \quad (n \rightarrow \infty) \quad (2.3)$$

uniformly for sufficiently small s .

We now consider the characteristic function of $p_n(k)$. For convenience we shift the mean by μ_n and the variance by σ_n^2 . Call this function $f_n(t)$. Since

$$\sum_k a_n(k) e^{ikx} = \varphi_n(ix),$$

it follows that

$$f_n(t) = e^{-i\mu_n t/\sigma_n} \varphi_n\left(\frac{it}{\sigma_n}\right) / \varphi_n(0).$$

From (2.3) we have uniformly for sufficiently small t/σ_n

$$\begin{aligned} f_n(t) &\sim e^{-i\mu_n t/\sigma_n} \left[\frac{r(0)}{r\left(\frac{it}{\sigma_n}\right)} \right]^n \\ &\sim \exp\{-i\mu_n t/\sigma_n - n(it\rho'(0)/\sigma_n - t^2\rho''(0)/2\sigma_n^2 + O(t^3/\sigma_n^3))\} \end{aligned} \quad (2.4)$$

by expanding $\rho(s) = \log r(s)/r(0)$ in a Taylor series. Setting $\sigma_n^2 = -n\rho''(0)$ and $\mu_n = -n\rho'(0)$, we obtain

$$f_n(t) \sim e^{-t^2/2}$$

for all t . By the continuity theorem [3, Section 13, Theorem 2] we have pointwise convergence of the distributions. As remarked in the introduction, this implies (1.1). Q.E.D.

3. APPLICATIONS

The easiest method for verifying (2.1) is usually to show that $f(z, e^s)$ is continuous for $s \leq \epsilon$ and z in the set

$$\{|z| \leq |r(0)| + \delta\} \cap \{|z - r(s)| \geq \eta\}$$

for some η . Since this is a compact set, f and hence (2.1) are bounded here. For $|z - r(s)| \leq \eta$ we can expand $f(z, e^s)$ in a Laurent series about $r(s)$ and show that the coefficient of the error term is bounded.

EXAMPLE 3.1. *Classical central limit theorem.* If $X_1, X_2, \dots$ are bounded, integer valued, independent, identically distributed random variables with $\text{prob}\{X_1 + X_2 + \dots + X_n = k\} = a_n(k)$, then

$$f(z, w) = \frac{1}{1 - z \sum_k a_1(k) w^k}.$$

Theorem 1 is easily shown to apply.

This is a special case of the following observation. Suppose

$$f(z, w) = \frac{g(z, w)}{P(z, w)^{m+1}} \quad (3.1)$$

where (i) $P(z, w)$ is a polynomial in z with coefficients continuous in w , (ii) $P(z, 1)$ has a simple root at r and all other roots have larger absolute value, (iii) $g(z, w)$ is analytic for w near 1 and $z < r + \epsilon$, and (iv) $g(r, 1) \neq 0$. Then the theorem applies. Many constant coefficient linear recursions are of this form.

Before proceeding it will be convenient to consider formulas for $r' = r'(s)$ and $r'' = r''(s)$. Let $r(s)$ be a root of $h(z, e^s) = 0$. We will take $h = P$ later. Differentiating we obtain

$$\begin{aligned} r'h_z + e^s h_w &= 0 \\ r''h_z + r'^2 h_{zz} + 2e^s r' h_{zw} + e^s h_w + e^{2s} h_{ww} &= 0. \end{aligned}$$

When $s = 0$ we obtain

$$\begin{aligned} r' &= -\frac{h_w}{h_z} \\ r'' &= -\frac{r'^2 h_{zz} + 2r' h_{zw} + h_w + h_{ww}}{h_z}. \end{aligned} \quad (3.2)$$

EXAMPLE 3.2. A dimer problem. McQuistan and Lichtman [7] studied $A(q, N)$, the number of ways to place q non-overlapping dimers on a $2 \times N$ array. The recursion

$$\begin{aligned} A(q, N) &= A(q, N-1) + 2A(q-1, N-1) + A(q-1, N-2) \\ &\quad - A(q-3, N-3) \end{aligned}$$

was proved. From this one easily derives

$$\sum_{q,N} A(q, N) w^q z^N = \frac{1 - wz}{1 - (z + 2wz + wz^2 - w^3 z^3)}.$$

They found that $r = 0.311108$ is the absolutely smallest root of $1 - 3z - z^2 + z^3$ and is simple. Applying Theorem 1 and (3.2) we find that $A(q, N)$ is asymptotically normal with

$$\begin{aligned} \mu &= 0.6065 \quad (\text{found by McQuistan and Lichtman}), \\ \sigma &= 0.4088. \end{aligned}$$

EXAMPLE 3.3. Separated samples. Let $a_n(k; m, c)$ be the number of k -tuples of integers $(i_1, \dots, i_k)$ such that $1 \leq i_1 < \dots < i_k \leq n$ and all but c of the $k-1$ differences $i_{j+1} - i_j$ are larger than m . We have [11]

$$\sum_{n=0}^{\infty} a_n(k; m, c) z^n = \binom{k-1}{c} \left(\frac{z^{m+1}}{1-z} \right)^{k-1-c} \frac{z^{c+1}(1-z^m)^c}{(1-z)^{c+2}}.$$

Hence

$$f(z, w) = \frac{z^{c+1}(1 - z^m)^c}{(1 - z)^{c+2}} \left(\frac{1}{1 - \frac{wz^{m+1}}{1 - z}} \right)^{c+1} w^{c+1}.$$

We are interested in $h(z, w) = 1 - z - wz^{m+1}$ since its absolutely smallest root is less than 1 for w near 1. Because $h_z(z, w) < 0$ for $z, w > 0$, this root will not be a multiple root.

Some computation using (3.2) yields

$$\mu = \frac{1 - r}{1 + m(1 - r)},$$

$$\sigma^2 = \frac{r(1 - r)}{(1 + m(1 - r))^3},$$

from which the following table was computed.

m	0	1	2	5	10	20	$\sim$
r	$\frac{1}{2}$	.618	.682	.778	.844	.897	$1 - (\log m)/m$
μ	$\frac{1}{2}$	.276	.194	.105	.061	.034	m^{-1}
σ	$\frac{1}{2}$	.299	.223	.136	.089	.057	$(\sqrt{m} \log m)^{-1}$

EXAMPLE 3.4. *Ordered set partitions.* Let $a_n(k)$ be the number of partitions of an n -set having exactly k labeled blocks, i.e., $k!S(n, k)$ where S is a Stirling number of the second kind. Harper showed that $S(n, k)$ is asymptotically normal. It is easily seen that

$$\sum_{n,k} a_n(k) \frac{z^n w^k}{n!} = \sum_k (e^z - 1)^k w^k$$

$$= \frac{1}{1 - w(e^z - 1)}.$$

There is a simple pole of $f(z, e^s)$ at $r(s) = \log(1 + e^{-s})$. Since

$$\frac{1}{1 - e^s(e^z - 1)} = \frac{1}{1 + e^s} \frac{1}{1 - e^{z-r(s)}}$$

and

$$(e^y - 1)^{-1} = \frac{1}{y} + O(1) \quad (3.3)$$

when y is bounded, the theorem applies. We have

$$\mu = \frac{1}{2 \log 2} = 0.72135,$$

$$\sigma = \frac{\sqrt{1 - \log 2}}{2 \log 2} = 0.39959.$$

EXAMPLE 3.5. *Eulerian numbers.* The Eulerian numbers $A_n(k)$ are the number of permutations of $1, 2, \dots, n$ having exactly k rises. It is well known that

$$\sum_{n,k} A_n(k) \frac{z^n w^k}{n!} = \frac{w(1-w)}{e^{(w-1)z} - w}. \quad (3.4)$$

This is extended to $w = 1$ by defining $f(z, 1) = (1 - z)^{-1}$. There is a simple pole at $r(s) = s(e^s - 1)^{-1}$. We have

$$\begin{aligned} f(z, e^s) &= \frac{1 - e^s}{e^{(e^s-1)(z-r(s))} - 1} \\ &= \frac{1 - e^s}{(e^s - 1)(z - r(s))} + O(1) \text{ by (3.3),} \end{aligned}$$

where the constant in O is bounded if s and $z - r(s)$ are small. Theorem 1 applies and we have

$$\mu = \frac{1}{2} \quad \sigma^2 = \frac{1}{12}.$$

David and Barton [2, pp. 150–154] derive (3.4) and then establish asymptotic normality of $A_n(k)$ by computing cumulants.

4. LOCAL LIMIT THEOREMS

It would be convenient to quote theorems in probability theory which would allow us to go from (1.1) to (1.2). Unfortunately, the type of theorems we need do not seem to exist. (The result of Moskvina, *et al.* [10] might be useful at times.) Consequently, this section is devoted to proving some local limit theorems. Theorem 2 is a straightforward local limit theorem. The local limit theorem is somewhat disguised in Theorems 3 and 4 since we are primarily interested in the asymptotic behavior of $a_n(k)$. These latter theorems are local analogs of Theorem 1.

Since

$$A \binom{n}{k} + (-1)^k B \binom{n}{k} \quad (4.1)$$

has the generating function

$$\frac{A}{1 - z(1 + w)} + \frac{B}{1 - z(1 - w)},$$

it is clear that the conditions in Theorem 1 are not enough. Theorem 4 imposes a smoothness condition on $a_n(k)$ directly while Theorem 3 does it indirectly by restricting $f(z, w)$. It should be possible to prove Theorem 4 from Theorem 3.

A sequence $a(k)$ is *unimodal* if it is first non-decreasing and then non-increasing.

LEMMA 1. *If (1.1) holds, $\sigma_n \rightarrow \infty$, and $p_n(k)$ is unimodal for sufficiently large n , then for every $\epsilon > 0$ the local limit theorem (1.2) holds for $S = \{x: |x| \geq \epsilon\}$. Furthermore, the rate of convergence depends only on ϵ , σ_n and the rate of convergence in (1.1).*

Proof. For $\delta > 0$ we have from (1.1)

$$\begin{aligned} \sum_{|k - \mu_n| \leq \delta \sigma_n} p_n(k) &\sim \frac{2}{\sqrt{2\pi}} \int_0^\delta e^{-t^2/2} dt \\ &> \frac{1}{\sqrt{2\pi}} \int_\delta^{3\delta} e^{-t^2/2} dt. \end{aligned}$$

Since this last integral is asymptotic to the sum of $p_n(k)$ over either of the intervals

$$\delta \sigma_n \leq k - \mu_n \leq 3\delta \sigma_n \quad (4.2)$$

or

$$\delta \sigma_n \leq \mu_n - k \leq 3\delta \sigma_n, \quad (4.3)$$

it follows that for sufficiently large n some value of $p_n(k)$ with $|k - \mu_n| < \delta \sigma_n$ is larger than some value of $p_n(k)$ with k in the range (4.2) and also with k in the range (4.3). By unimodality, $p_n(k)$ is non-increasing for $k > \mu_n + \delta \sigma_n$ for sufficiently large n and likewise non-decreasing for $k < \mu_n - \delta \sigma_n$.

Choose $x \in S$. We may assume $x > 0$. For $0 < \gamma < \epsilon$ define $l = [\gamma\sigma_n]$ and $x_n = [x\sigma_n + \mu_n]$. When n is sufficiently large

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int_{x-\gamma}^x e^{-t^2/2} dt &\sim \sum_{k=x_n-l-1}^{x_n} p_n(k) \\ &\geq lp_n(x_n) \\ &\geq \sum_{k=x_n}^{x_n+l} p_n(k) \\ &\sim \frac{1}{\sqrt{2\pi}} \int_x^{x+\gamma} e^{-t^2/2} dt. \end{aligned}$$

By (1.1) this convergence is uniform. Hence we have

$$lp_n(x_n) \sim \frac{\gamma}{\sqrt{2\pi}} e^{-x^2/2 + O(x\gamma)}$$

and so for some c

$$\left| \sigma_n p_n(x_n) - \frac{e^{-x^2/2}}{\sqrt{2\pi}} \right| \leq \frac{e^{-x^2/2}}{\sqrt{2\pi}} (e^{e|\gamma x|} - 1).$$

The right hand side is $o(1)$ as $\gamma \rightarrow 0$.

Q.E.D.

The lemma cannot be extended to cover the region near μ_n because a unimodal function can behave poorly there. A sufficient condition is *log concavity*:

$$(a_n(k))^2 \geq a_n(k+1) a_n(k-1).$$

This is also known as 2-positivity in the literature.

LEMMA 2. *If $a_n(k)$ is log concave and satisfies (1.1) and if $\sigma_n \rightarrow \infty$, then (1.2) holds for $S = (-\infty, +\infty)$. Furthermore, the rate of convergence depends only on σ_n and the rate of convergence in (1.1).*

Proof. Log concavity of $a_n(k)$ states precisely that $\log a_n(k)$ is concave. Hence $p_n(k)$ is unimodal and

$$p_n(j+l)^2 \geq p_n(j) p_n(j+2l). \quad (4.4)$$

By Lemma 1 and symmetry we may assume $0 \leq x \leq \epsilon$. For $\gamma > 0$ let $l = [\gamma\sigma_n]$ and $j = [x\sigma_n + \mu_n]$.

By (4.4)

$$\begin{aligned}\sigma_n p_n(j) &\leq \sigma_n p_n(j+l) \frac{p_n(j+l)}{p_n(j+2l)} \\ &= \frac{e^{-x^2/2}}{\sqrt{2\pi}} (1 + O(\gamma))\end{aligned}$$

by Lemma 1. On the other hand, by the proof of Lemma 1, $p_n(j) \geq p_n(j+l)$ for sufficiently large n . Hence

$$\begin{aligned}\sigma_n p_n(j) &\geq \sigma_n p_n(j+l) \\ &= \frac{e^{-x^2/2}}{\sqrt{2\pi}} (1 + O(\gamma)).\end{aligned}$$

Thus the left-hand side of (1.2) is $O(\gamma)$. Since γ is arbitrary, we are done. Q.E.D.

THEOREM 2 (cf. Harper). *Suppose $P_n(x) = \sum_k a_n(k) x^k$ is a polynomial whose roots are all real and non-positive. If $\sigma_n \rightarrow \infty$, then (1.2) holds for $S = (-\infty, +\infty)$. In fact*

$$\begin{aligned}\mu_n &= \sum_j \frac{1}{1 + r_n(j)}, \\ \sigma_n^2 &= \sum_j \frac{r_n(j)}{(1 + r_n(j))^2},\end{aligned}$$

where

$$P_n(x) = a_n \prod_j (x + r_n(j)).$$

Proof. Define independent random variables X_{nj} by

$$\begin{aligned}\text{prob}\{X_{nj} = 0\} &= \frac{r_n(j)}{1 + r_n(j)}, \\ \text{prob}\{X_{nj} = 1\} &= \frac{1}{1 + r_n(j)}.\end{aligned}$$

Then

$$p_n(k) = \text{prob} \left\{ \sum_j X_{nj} = k \right\}.$$

By the general central limit theorem [3, Section 26, Theorem 2] with

$\xi_{nj} = X_{nj}/\sigma_n$, we see that (1.1) holds if $\sigma_n \rightarrow \infty$. By Newton's inequality (see Lieb [6]), it follows that the $p_n(k)$ are log concave. Apply Lemma 2. Q.E.D.

THEOREM 3. *Let $f(z, w)$ have a power series expansion*

$$f(z, w) = \sum_{n,k \geq 0} a_n(k) z^n w^k$$

with non-negative coefficients and let $a < b$ be real numbers. Define

$$R(\epsilon) = \{z: a \leq \operatorname{Re} z \leq b, |\operatorname{Im} z| \leq \epsilon\}.$$

Suppose there exists $\epsilon > 0$, $\delta > 0$, a non-negative integer m , and functions $A(s), r(s)$ such that

- (i) $A(s)$ is continuous and non-zero for $s \in R(\epsilon)$;
- (ii) $r(s)$ is non-zero and has a bounded third derivative for $s \in R(\epsilon)$;
- (iii) for $s \in R(\epsilon)$ and $|z| \leq |r(s)|(1 + \delta)$

$$\left(1 - \frac{z}{r(s)}\right)^m f(z, e^s) - \frac{A(s)}{1 - z/r(s)}$$

is analytic and bounded;

(iv)

$$\left(\frac{r'(\alpha)}{r(\alpha)}\right)^2 - \frac{r''(\alpha)}{r(\alpha)} \neq 0 \quad \text{for } a \leq \alpha \leq b;$$

(v) $f(z, e^s)$ is analytic and bounded for

$$|z| \leq |r(\operatorname{Re} s)|(1 + \delta) \quad \text{and} \quad \epsilon \leq |\operatorname{Im} s| \leq \pi.$$

Then we have

$$a_n(k) \sim \frac{n^m e^{-\alpha k} A(\alpha)}{m! r(\alpha)^n \sigma_\alpha \sqrt{2\pi n}} \quad (4.6)$$

uniformly for $a \leq \alpha \leq b$, where

$$\begin{aligned} \frac{k}{n} &= -\frac{r'(\alpha)}{r(\alpha)}, \\ \sigma_\alpha^2 &= \left(\frac{k}{n}\right)^2 - \frac{r''(\alpha)}{r(\alpha)}. \end{aligned} \quad (4.7)$$

Proof. We follow the idea used in [3, Section 49] for identically distributed, independent, random variables. We need to study $f_n(t)$ —the

characteristic function for the sequence $e^{\alpha k} a_n(k)$ adjusted to have mean 0 and variance 1. From the proof of Theorem 1 we have

$$f_n(t) = e^{-i\mu_n t/\sigma_n} \varphi_n\left(\alpha + \frac{it}{\sigma_n}\right) / \varphi_n(\alpha),$$

where by (2.3)

$$\begin{aligned} \varphi_n(s) &= \sum_k e^{sk} a_n(k) \\ &= \binom{n+m}{n} r(s)^{-n} A(s) \left(1 + O\left(\frac{1}{n}\right)\right), \end{aligned}$$

the O being uniform on $R(\epsilon)$. By (2.4) we have uniformly for $a \leq \alpha \leq b$

$$f_n(t) = e^{-t^2/2(1+O(t/\sigma_n))} \left(1 + O\left(\frac{1}{n}\right)\right) \frac{A(\alpha + it/\sigma_n)}{A(\alpha)}. \quad (4.8)$$

Now we have (see [3, Section 49]) for $z = z_{nk} = (k - \mu_n)/\sigma_n$, $B > 0$ fixed, and $\epsilon > 0$ small and fixed

$$\begin{aligned} 2\pi \left| \sigma_n p_n(k) - \frac{e^{-z^2/2}}{\sqrt{2\pi}} \right| &= \left| \int_{-\pi\sigma_n}^{\pi\sigma_n} e^{-izx} f_n(x) dx - \int_{-\infty}^{+\infty} e^{-izx - x^2/2} dx \right| \\ &= |I_1 + I_2 + I_3 + I_4| \\ &\leq |I_1| + |I_2| + |I_3| + |I_4|, \end{aligned}$$

where

$$I_1 = \int_{-B}^B e^{-izx} (f_n(x) - e^{-x^2/2}) dx,$$

$$I_2 = \int_{B \leq |x| \leq \epsilon\sigma_n} e^{-izx} f_n(x) dx,$$

$$I_3 = \int_{\epsilon\sigma_n \leq |x| \leq \pi\sigma_n} e^{-izx} f_n(x) dx,$$

$$I_4 = \int_{B \leq |x|} e^{-izx - x^2/2} dx.$$

By (4.8) we have

$$|I_1| \leq 2 \int_0^B e^{-x^2/2} O\left(\frac{B^3}{\sigma_n}\right) dx = o(1).$$

Again by (4.8)

$$\begin{aligned} |I_2| &\leq 2 \int_B^{\epsilon \sigma_n} e^{-x^2/2(1+O(\epsilon))} (1 + O(\epsilon)) dx \\ &\leq 3 \int_B^{\infty} e^{-x^2/4} dx \quad \text{when} \quad |O(\epsilon)| \leq \frac{1}{2} \\ &= O\left(\frac{1}{B}\right), \end{aligned}$$

since

$$\int_B^{\infty} e^{-cx^2} dx = O(B^{-1}) \quad \text{for } c > 0.$$

Also $|I_4| = O(B^{-1})$. Finally we consider I_3 . By (v) we have

$$\left| \sum_k e^{sk} a_n(k) \right| = \left| \frac{1}{2\pi i} \oint_{|z|=r} \frac{f(z, e^s)}{z^{n+1}} dz \right| \leq Kr^{-n},$$

where $s = \alpha + iy$, $\epsilon \leq |y| \leq \pi$, and $r = r(\alpha)(1 + \delta)$. Hence

$$\begin{aligned} |I_3| &\leq \int_{\epsilon \sigma_n \leq |x| \leq \pi \sigma_n} |f_n(x)| dx \\ &= \sigma_n \int_{\epsilon \leq |y| \leq \pi} \frac{\sum_k e^{iyk} e^{\alpha k} a_n(k)}{\sum_k e^{\alpha k} a_n(k)} dy \\ &= \sigma_n O\left(\left(\frac{r(\alpha)}{r}\right)^n\right) \\ &= o(1). \end{aligned}$$

Hence

$$\left| \sigma_n p_n(k) - \frac{e^{-z^2/2}}{\sqrt{2\pi}} \right| = O\left(\frac{1}{B}\right) + o(1)$$

for every $B > 0$ and so the left-hand side tends to zero as $n \rightarrow \infty$.

We now set $z = 0$. This means $\mu_n = k/n$, where μ_n and σ_n are given by Theorem 1. Combining this with (2.3) we obtain (4.6) and (4.7). Q.E.D.

Condition (v) in Theorem 3 is the essential condition which allows us to go from a central limit theorem to a local limit theorem. Since $a_n(k) \geq 0$, it follows that

$$\min_y |r(\alpha + iy)| = r(\alpha), \quad \alpha \text{ real.}$$

However, we want the minimum to be uniquely achieved for each α to avoid behavior like (4.1).

It is possible to replace condition (v) by unimodality:

THEOREM 4. *If condition (v) in Theorem 3 is replaced by the assumption (v') $e^{\alpha k} a_n(k)$ is unimodal for sufficiently large n and $a \leq \alpha \leq b$, then the conclusion is still valid. Condition (v') is implied by (v'') $a_n(k)$ is log concave for sufficiently large n .*

Proof. That (v'') implies (v') is clear so we consider (v'). To begin, we wish to show that convergence to the central limit theorem is uniform for $a \leq \alpha \leq b$. Apply Theorem 1 of [3, Section 39] with $G(x)$ the normal distribution, $F(x) = F_n(x)$ the distribution for $e^{\alpha k} a_n(k)$, and $T = n^{1/8}$. One easily obtains that

$$\int_{-T}^T \left| \frac{f_n(t) - g(t)}{t} \right| dt$$

is uniformly bounded because of (4.8).

For fixed ϵ , the convergence in Lemma 1 is uniform for $a \leq \alpha \leq b$. The idea now is to consider α_τ defined by

$$\frac{k + \tau \sigma_\alpha \sqrt{n}}{n} = - \frac{r'(\alpha_\tau)}{r(\alpha_\tau)}. \quad (4.9)$$

For fixed $\tau \neq 0$ we have

$$a_n(k) \sim \frac{n^m e^{-k\alpha_\tau} A(\alpha_\tau)}{m! r(\alpha_\tau)^n \sigma_{\alpha_\tau} \sqrt{2\pi n}} e^{-\tau^2/2}$$

uniformly by Lemma 1. We study this for small τ . Clearly

$$a_n(k) \sim \frac{n^m e^{-k\alpha} A(\alpha)}{m! r(\alpha)^n \sigma_\alpha \sqrt{2\pi n}} e^u, \quad (4.10)$$

where, setting $\rho = \log r$,

$$u = n(\rho(\alpha) - \rho(\alpha_\tau)) + k(\alpha - \alpha_\tau) - \tau^2/2.$$

From (4.9)

$$\alpha - \alpha_\tau = \frac{\tau \sigma_\alpha}{\rho''(\alpha) \sqrt{n}} + O(n^{-1}).$$

Expanding $\rho(\alpha_\tau)$ about α and using $\sigma_\alpha^2 = -\rho''(\alpha)$ we obtain $u = O(n^{-1/2})$. Since $A(\alpha)$, σ_α , and α_τ are continuous and $a_n(k)$ is independent of τ , we obtain (4.6) from (4.10). Q.E.D.

5. APPLICATIONS

We now give some applications of the theorems just proved.

EXAMPLE 5.1. *Stirling numbers of the first kind.* Let $s(n, k)$ be the signless Stirling numbers of the first kind, defined by

$$P_n(x) = \sum_k s(n, k) x^k = x(x+1) \cdots (x+n-1).$$

We wish to apply Theorem 2 to $P_n(\alpha x)$. The mean and variance are given by

$$\mu_n = \sum_{i=0}^{n-1} \frac{\alpha}{\alpha+i} \approx \int_0^n \frac{\alpha dx}{\alpha+x} = \alpha \log \left(1 + \frac{n}{\alpha}\right). \quad (5.1)$$

$$\sigma_n^2 = \sum_{i=0}^{n-1} \frac{\alpha i}{(\alpha+i)^2} \approx \int_0^n \frac{\alpha x dx}{(\alpha+x)^2} = \alpha \log \left(1 + \frac{n}{\alpha}\right) - \frac{\alpha n}{\alpha+n}.$$

We must decide when $\sigma_n \rightarrow \infty$. If α is small compared with n we have

$$\sigma_n^2 \approx \alpha \log \left[\frac{1}{\alpha} (n + \alpha) \right] - \frac{\alpha}{1 + \alpha/n} \approx \alpha \log n.$$

When α is large compared with n ,

$$\sigma_n^2 \approx n - \frac{n^2}{2\alpha} - \frac{n}{1 + \frac{n}{\alpha}} \approx \frac{n^2}{2\alpha}.$$

In the intermediate range $\sigma_n^2 \rightarrow \infty$. From the above we see that $\sigma_n^2 \rightarrow \infty$ provided

$$\alpha \log n \rightarrow \infty \quad \text{and} \quad \frac{\alpha}{n^2} \rightarrow 0.$$

By (5.1) this is equivalent to

$$\mu_n \rightarrow \infty \quad \text{and} \quad n - \mu_n \rightarrow \infty.$$

Since $P_n(x) = \Gamma(x+n)/\Gamma(x)$ we have

$$s(n, k) \sim \frac{\Gamma(\alpha+n)}{\alpha^k \Gamma(\alpha) \sigma \sqrt{2\pi}} \quad (5.2)$$

as $k, n - k \rightarrow \infty$ simultaneously where $\alpha = \alpha(n, k)$ and $\sigma = \sigma(n, k)$ are determined by

$$\begin{aligned}\frac{k}{n} &= \sum_{i=0}^{n-1} \frac{\alpha}{\alpha + i}, \\ \sigma^2 &= \sum_{i=0}^{n-1} \frac{\alpha i}{(\alpha + i)^2}.\end{aligned}\tag{5.3}$$

This result is the first term of an asymptotic expansion due to Moser and Wyman [8]. They obtained their result by a detailed study of the generating function. The range of validity we have established is somewhat larger than they gave.

EXAMPLE 5.2. Separated samples. The notation is as in Example 3.3. We will apply Theorem 3. The poles of $f(z, w)$ with $|z| < 1$ are given by $1 - z - wz^{m+1} = 0$. Let $w = e^\alpha e^{i\theta}$. The equation

$$1 = z(1 + e^\alpha z^m)$$

has a unique root $r(\alpha)$ of minimum modulus and it lies between 0 and 1 —the arguments of Example 3.3 apply. Suppose $0 < |\theta| \leq \pi$, then for $|z| \leq r(\alpha)$

$$|z(1 + e^\alpha e^{i\theta} z^m)| < r(\alpha)(1 + e^\alpha r(\alpha)^m) = 1$$

unless $z^m = r(\alpha)^m e^{-i\theta}$. In this case

$$z(1 + e^\alpha z^m) = e^{-i\theta/m} \neq 1.$$

Hence all roots of $1 - z - e^\alpha e^{i\theta} z^{m+1}$ exceed $r(\alpha)$ in modulus for $0 < |\theta| \leq \pi$. Using (3.2) for

$$h(z, w) = we^\alpha z^{m+1} + z - 1$$

and using $h(z, 1) = 0$ we find after calculation that

$$\begin{aligned}\frac{k}{n} &= \frac{1 - r}{1 + m(1 - r)}, \\ \sigma_\alpha^2 &= \frac{r(1 - r)}{(1 + m(1 - r))^3} = \left(\frac{k}{n}\right)^3 \frac{r}{(1 - r)^2}.\end{aligned}\tag{5.4}$$

Since r ranges over $(0, 1)$ as α ranges over $(-\infty, +\infty)$, it follows from (5.4) that k/n ranges over $(0, (m+1)^{-1})$. Hence Theorem 3 applies for $c_1 \leq k/n \leq c_2$ where $c_1 > 0$ and $c_2 < (m+1)^{-1}$.

Since

$$f(z, e^\alpha) = \left(\frac{e^\alpha z(1 - z^m)}{1 - z - e^\alpha z^{m+1}} \right)^{c+1} \frac{1}{(1 - z)(1 - z^m)},$$

the pole at $z = r$ is of order $c + 1$. Since

$$\lim_{z \rightarrow r} (1 - z/r)(1 - z - e^\alpha z^{m+1})^{-1} = (rh_z(r, 1))^{-1},$$

we have

$$A(\alpha) = \left(\frac{(e^\alpha + 1)r - 1}{n(1 - r)} k \right)^{c+1} \frac{re^\alpha}{((e^\alpha + 1)r - 1)(1 - r)}. \quad (5.5)$$

By (4.6), (5.4), and (5.5) we have

$$a_n(k; m, c) \sim \frac{k^{c-1/2} e^{\alpha(1-k)}}{c! \sqrt{2\pi} r^{n-1/2} (1 - r)} \left(\frac{(e^\alpha + 1)r - 1}{1 - r} \right)^c \quad (5.6)$$

uniformly as $n \rightarrow \infty$, provided $c_1 \leq k/n \leq c_2$, where $c_1 > 0$ and $c_2 < (m + 1)^{-1}$ are arbitrary constants,

$$r = 1 - \frac{k}{n - km} \quad \text{and} \quad e^\alpha = \frac{1 - r}{r^{m+1}}.$$

EXAMPLE 5.3. *Eulerian numbers.* Theorem 3 can be used here as in the previous example. From Example 3.5

$$f(z, w) = \frac{w(1 - w)}{e^{(w-1)z} - w},$$

when $a_n(k) = A_n(k)/n!$. Hence $r(s) = s(e^s - 1)^{-1}$ and

$$\left| \frac{r(\alpha + iy)}{r(\alpha)} \right|^2 = \frac{1 + (y/\alpha)^2}{1 + 2(1 - \cos y)(e^{\alpha/2} - e^{-\alpha/2})^{-2}}.$$

This exceeds 1 if and only if

$$(y/\alpha)^2 > 2(1 - \cos y)(e^{\alpha/2} - e^{-\alpha/2})^{-2},$$

which is valid since

$$y^2 > 2(1 - \cos y) \quad \text{and} \quad |e^{\alpha/2} - e^{-\alpha/2}| \geq |\alpha|$$

for $y \neq 0$. Condition (iii) is verified as in Example 3.5. Applying Theorem 3 we have

$$\begin{aligned} A(\alpha) &= \lim_{z \rightarrow r} \frac{(1 - e^\alpha)(1 - z/r)}{e^{(e^\alpha - 1)(z - r)} - 1}; \quad r = \frac{\alpha}{e^\alpha - 1} \\ &= r^{-1}. \end{aligned}$$

Hence for every $\epsilon > 0$

$$A_n(k) \sim \frac{n! e^{-\alpha k}}{r^{n+1} \sigma_\alpha \sqrt{2\pi n}} \quad (5.7)$$

uniformly for $\epsilon \leq k/n \leq 1 - \epsilon$ where α , r and σ_α are determined by

$$\begin{aligned} \frac{k}{n} &= \frac{e^\alpha}{e^\alpha - 1} - \frac{1}{\alpha}, \\ r &= \frac{\alpha}{e^\alpha - 1}, \\ \sigma_\alpha^2 &= \frac{1}{\alpha^2} - \frac{e^\alpha}{(e^\alpha - 1)^2}. \end{aligned} \quad (5.8)$$

The range for k/n follows from (5.8) since

$$\lim_{\alpha \rightarrow -\infty} k/n = 0 \quad \text{and} \quad \lim_{\alpha \rightarrow +\infty} k/n = 1.$$

Frobenius [1, p. 250] observed that $\sum A_n(k) x^k$ has non-positive roots. By Newton's inequality [5], the $A_n(k)$ are log concave. Hence we could derive (5.7) by means of Theorem 4.

EXAMPLE 5.4. *Stirling numbers of the second kind.* Here there are three formulas from which to choose:

- (a) $\sum_k S(n, k) x^k$ has non-positive, real roots (Harper);
- (b) $\sum_k S(n, k) k! \binom{n}{k} x^k = \sum_k x^k (1-x)^{n-k} k^n \binom{n}{k}$ has non-positive, real roots (Lieb);
- (c) $\sum_{n,k} k! S(n, k) \frac{z^n}{n!} w^k = (1 - w(e^z - 1))^{-1}$.

In both (a) and (b) determination of the mean and variance is difficult. Consequently we turn to (c). The pole of (c) is at

$$z = \log(1 + e^{-s}).$$

Since $e^z - 1$ has a Taylor series with non-negative coefficients, $1 = w(e^z - 1)$ can be dealt with by the method of Example 5.2. Also, Lieb's inequality

$$S(n, k)^2 \geq S(n, k-1) S(n, k+1) \frac{k}{k-1}$$

implies that $k! S(n, k)$ is log concave. By Theorem 3 or 4 we have, after some calculation, for every $\epsilon > 0$

$$S(n, k) \sim \frac{n! e^{-\alpha k}}{k! r^{n+1} (e^\alpha + 1) \sigma_\alpha \sqrt{2\pi n}} \quad (5.9)$$

uniformly for $\epsilon < k/n < 1 - \epsilon$ where α , r and σ_α are determined by

$$\begin{aligned} \frac{k}{n} &= \frac{1}{(1 + e^\alpha) \log(1 + e^{-\alpha})}, \\ r &= \log(1 + e^{-\alpha}), \\ \sigma_\alpha^2 &= \left(\frac{k}{n}\right)^2 (1 - e^\alpha \log(1 + e^{-\alpha})). \end{aligned} \quad (5.10)$$

The range for k/n is established by the fact that formula (5.10) for k/n ranges over $(0, 1)$ as α ranges over $(-\infty, +\infty)$. Equation (5.9) is the first term of an asymptotic expansion obtained by Laplace [2, p. 312]. Moser and Wyman [9] also discuss the problem. Their result is valid for a somewhat larger range of k/n .

EXAMPLE 5.5. Inversions in a permutation. Let $a_1, a_2, \dots, a_n$ be a permutation of $1, 2, \dots, n$. A pair (i, j) is an *inversion* if $i < j$ and $a_i > a_j$. Let $a_n(k)$ be the number of permutations of $1, 2, \dots, n$ having exactly k inversions. By considering the i for which $a_i = n + 1$, it is easily shown that

$$a_{n+1}(k) = \sum_{j=0}^n a_n(k - j) \quad (5.11)$$

and so

$$P_n(x) = \sum_k a_n(k) x^k = \prod_{k=1}^n \frac{1 - x^k}{1 - x}. \quad (5.12)$$

Unfortunately, the theorems of Section 4 do not apply; however, the spirit does. If we had a local limit theorem, then we would have

$$a_n(k) \sim \frac{P_n(\alpha) \alpha^{-k}}{\sigma \sqrt{2\pi}}, \quad (5.13)$$

where

$$\begin{aligned} k &= \frac{d \log P_n(\alpha)}{d \log \alpha}, \\ \sigma^2 &= \frac{d^2 \log P_n(\alpha)}{(d \log \alpha)^2}. \end{aligned} \quad (5.14)$$

It is easily seen from (5.12) that $a_n(k) = a_n(\binom{n}{2} - k)$ and it is not hard to show that (5.13) produces identical estimates for $a_n(k)$ and $a_n(\binom{n}{2} - k)$.

As in Theorem 2, we can introduce independent random variables associated with each of the factors $(1 - x^k)/(1 - x)$. We want to replace $a_n(k)$ by $\alpha^k a_n(k)$ to shift the mean, where α depends on n . Hence

$$\text{prob}\{X_{nk} = i\} = \begin{cases} \alpha^i \left(\sum_{j=0}^{k-1} \alpha^j \right)^{-1}, & 0 \leq i < k, \\ 0, & \text{otherwise.} \end{cases}$$

If $\alpha = 1$, then the variance of X_{nk} is asymptotic to $k^2/12$ for large n and k . On the other hand, if α is bounded away from 1, the variance of X_{nk} approaches $\alpha/(1 - \alpha)^2$. Thus there are two different sorts of behavior. By the central limit theorem [3, Section 26, Theorem 2] it is easy to show that $\alpha^k a_n(k)$ satisfy (1.1) provided $\alpha = 1$ or $|1 - \alpha| \geq c$ and $n\alpha(1 - \alpha)^{-2} \rightarrow \infty$ where $c > 0$ is arbitrary. Extension to the case $\alpha \rightarrow 1$ should be possible, but it appears to be somewhat tricky.

We will show that $a_n(k)$ is log concave so that Lemma 2 applies. By (5.11)

$$\begin{aligned} a_{n+1}(k)^2 &= a_{n+1}(k-1) a_{n+1}(k+1) \\ &+ \sum_{j=0}^n [a_n(k-j) a_n(k-n) - a_n(k+1-j) a_n(k-n-1)] \\ &+ \sum_{j=0}^n [a_n(k+1-j) a_n(k) - a_n(k-j) a_n(k+1)]. \end{aligned}$$

If we assume the $a_n(k)$ are log concave, then each bracketed term in non-negative and so the $a_{n+1}(k)$ are log concave. Since $a_1(k) = \delta_{0k}$ they are log concave.

6. DISCUSSION

As noted in Section 1, when Theorem 1 applies we can expect a result like (4.6) unless something obvious is wrong as in (4.1). Such heuristic results could be useful in two ways. From a theoretical point of view, they can lead to formulas which could be established and perhaps improved by using other methods. From a practical point of view, they give results which could be tested for accuracy using small (to a digital computer) values of n . One could then boldly use them in the untested range.

It seems reasonable on the basis of experience to expect many linear homogeneous recursions to lead to asymptotically normal $a_n(k)$. If the coefficients are not constant, some assumptions about them will be needed since

$$a_n(k) = ((n - k)^2 + k^2)(a_{n-1}(k) + a_{n-1}(k - 1))$$

leads to a bimodal distribution. Our tools seem inappropriate for attacking the general problem. In the case of constant coefficients it might be possible to apply Theorem 3 to obtain a fairly general theorem since $f(z, w)$ is usually rational. If the denominator happens to have the form $1 - R(z, w)$ where R has positive coefficients and no constant term, a procedure like that in Example 5.2 can often be used. If R has the form $zg(w)$, then

$$f(z, w) = \sum g(w)^n z^n$$

and $g(w)^n$ can be studied by the saddle point method. See Good [4] and Moser and Wyman [9].

Note added in proof. For linear homogeneous recursions generated by the method of Read (Contributions to the cell growth problem, *Canad. J. Math.* **14** (1962), 1–20, Theorem 1) we can show that Theorem 3 applies.

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